Improving Artificial Teachers by Considering How People Learn and Forget

Submitted by

Gaurang Kishorbhai Sumara

Master of Science

in

Data Science

Hamburg University of Technology

Human-Centric Machine Learning

Supervisor:

Prof. Dr. Pierre-Alexandre Murena

March 6, 2026

Abstract

Human memory does not fail randomly. Forgetting follows systematic patterns — accelerating with time while slowing with repeated exposure — and these regularities make computational modelling of memory both feasible and practically valuable. Accurate modelling of these dynamics is essential for developing artificial teaching systems that optimize review schedules and improve long-term retention.

This project presents a hierarchical Bayesian model of human memory that captures both user-specific and item-specific variation in forgetting behaviour. The model is based on an exponential memory decay formulation embedded within a probabilistic framework that supports uncertainty quantification and hierarchical parameter sharing. Posterior inference is performed using Markov Chain Monte Carlo methods, specifically the No-U-Turn Sampler.

Posterior analysis indicates that the hierarchical formulation produces stable parameter estimates and captures meaningful variability across learners and learning items. By maintaining probabilistic representations of memory parameters rather than fixed point estimates, the model provides a principled basis for uncertainty-aware prediction of recall probability and supports the development of personalised artificial teaching systems.

Contents

Abstract **1**

1 Introduction **5**

1.1 Background 5

1.2 Problem Statement 5

1.3 Research Objective 6

1.4 Contributions 6

2 Related Work **7**

2.1 Classical Models of Human Memory 7

2.2 Memory Models in Artificial Teaching Systems 8

2.3 Bayesian Approaches in Memory Modeling 8

3 Mathematical Model **10**

3.1 Overview 10

3.2 Memory Decay Model 10

3.3 Observation Model 11

3.4 Frequentist Parameter Estimation 11

4 Bayesian Hierarchical Model **13**

4.1 Motivation for Bayesian Modeling 13

4.2 Bayesian Formulation 13

4.3 Prior Specification 14

4.3.1 Initial Prior Specification 14

4.3.2 Reparameterization for Efficient Inference 15

4.3.3 Forgetting Rate 15

4.3.4 Repetition Effect . . . 15
4.4 Hierarchical Structure . . . 16
4.4.1 Graphical Model Representation . . . 16
4.4.2 User-Level Parameters . . . 17
4.4.3 Item-Level Parameters . . . 18
4.4.4 Interaction-Level Parameters . . . 18
4.4.5 Generative Model . . . 18
4.5 Non-Centered Parameterization . . . 19
4.6 Posterior Inference Objective . . . 19

5 Posterior Inference and Sampling **20**
5.1 Inference Objective . . . 20
5.2 Markov Chain Monte Carlo Methods . . . 20
5.3 Hamiltonian Monte Carlo . . . 21
5.4 No U-Turn Sampler (NUTS) . . . 22
5.5 Posterior Geometry in Hierarchical Models . . . 22
5.6 The Funnel Problem in Memory Modeling . . . 22
5.7 Non-Centered Parameterization . . . 23
5.8 Sampling Diagnostics . . . 23
5.9 Posterior Interpretation . . . 24

6 Implementation **25**
6.1 System Overview . . . 25
6.2 Software Framework . . . 26
6.3 Data Representation . . . 27
6.4 Model Construction . . . 27
6.5 Parametrization Strategy . . . 28
6.6 Posterior Inference Pipeline . . . 28
6.7 Simulation and Training Workflow . . . 29
6.8 Computational Considerations . . . 29

7 Posterior Analysis and Model Behavior **30**
7.1 Overview . . . 30
7.2 Experimental Setup . . . 31

7.3 Posterior Distribution of User Parameters 31
7.4 Sampling Behaviour and Convergence Diagnostics 32
7.5 Variability Across Learners 33
7.6 Joint Posterior Structure 34
7.7 Interpretation for Artificial Teaching Systems 35

8 Discussion, Limitations, and Future Directions 37

8.1 Discussion 37
8.2 Limitations 37
8.3 Conclusion 38
8.4 Future Directions 39
8.4.1 Evaluation on Real Learning Data 39
8.4.2 Scalable Approximate Inference 39
8.4.3 Richer Cognitive Models 39
8.4.4 Integration with Intelligent Tutoring Systems 39

Chapter 1

Introduction

1.1 Background

Human memory plays a central role in learning and knowledge retention. However, memory is not static and gradually decays over time if information is not reinforced. This phenomenon, known as forgetting, has been extensively studied in cognitive science. Understanding and modeling memory dynamics is essential for designing effective learning systems.

Artificial teaching systems, such as spaced repetition platforms, aim to optimize the timing of review sessions in order to maximize long-term retention. These systems rely on mathematical models of memory to predict recall probability and schedule reviews accordingly.

Traditional memory models often use fixed parameters and do not account for differences between individuals or learning materials. However, in practice, memory behavior varies significantly across users and items.

1.2 Problem Statement

Existing memory models have several limitations. Various models estimate parameters using frequentist methods, such as maximum likelihood estimation(MLE), which provide point estimates without quantifying uncertainty. These approaches can lead to unstable parameter estimates, particularly when data is limited.

Furthermore, traditional models do not adequately capture hierarchical structure,

such as variation between users and items. This limits their ability to provide personalized predictions.

1.3 Research Objective

The objective of this project is to develop a *hierarchical Bayesian model* of human memory that captures individual differences in forgetting behavior while quantifying uncertainty in parameter estimates. The model aims to provide a probabilistic framework for predicting recall probability and improving artificial teaching systems.

1.4 Contributions

The main contributions of this project are:

- Development of a probabilistic memory model based on exponential forgetting dynamics,
- Extension of the model using hierarchical Bayesian formulation,
- Implementation of efficient inference using Markov Chain Monte Carlo methods,
- Modeling of user-specific and item-specific memory parameters,
- Analysis and interpretation of posterior parameter estimates.

Chapter 2

Related Work

2.1 Classical Models of Human Memory

The systematic study of human forgetting originates with the seminal work of Ebbinghaus [2]. Through controlled memorisation experiments, he demonstrated that memory retention declines predictably over time, a phenomenon now widely known as the forgetting curve. This observation provided the first empirical evidence that forgetting is not random but follows regular, quantifiable patterns.

A commonly used mathematical representation of this phenomenon is the exponential decay model, in which the probability of recall decreases exponentially with the time elapsed since the last review [9].Despite its simplicity, this formulation has proven to be a useful approximation of memory dynamics in both psychological research and educational applications [2] [9].

An alternative line of work proposes power-law models of forgetting, which predict that memory declines rapidly at first but slows over longer time scales [1] [11]. Although such models can sometimes achieve better empirical fit, they are often more difficult to interpret and integrate into practical learning systems [9].

Despite their contributions, classical memory models typically assume that forgetting parameters are fixed and identical across learners and learning materials [5]. In practice, however, substantial variability exists: different learners forget at different rates, and different items vary considerably in difficulty and memorability. Addressing this heterogeneity is therefore essential for developing realistic computational models of human memory.

2.2 Memory Models in Artificial Teaching Systems

Artificial teaching systems, particularly spaced repetition platforms rely on mathematical models of memory to determine when learning materials should be reviewed [8]. The goal of such systems is to present information at the point where the probability of forgetting becomes sufficiently high, thereby strengthening memory retention while avoiding unnecessary repetitions [5].

Many deployed systems employ heuristic scheduling rules or simple parametric models whose parameters are estimated using frequentist methods such as maximum likelihood estimation [9]. For example, the SuperMemo algorithm [11] determines inter-repetition intervals based on empirically tuned rules derived from observed recall performance. Although such approaches are computationally efficient and widely used in practice, they typically produce only point estimates of memory parameters and provide no mechanism for quantifying uncertainty in the underlying model.

A further limitation of many existing systems is the assumption that memory parameters are fixed and shared across all learners and learning items [8] [5]. In practice, however, learners differ substantially in their memory abilities and learning histories, while items vary in intrinsic difficulty and memorability. Ignoring these sources of variability can reduce the precision of review scheduling and limit the effectiveness of personalised learning systems.

These limitations motivate the development of probabilistic memory models capable of capturing uncertainty and representing variability across both learners and items.

2.3 Bayesian Approaches in Memory Modeling

Bayesian inference provides a principled framework for representing uncertainty in cognitive models [3]. Rather than producing single point estimates of model parameters, Bayesian methods infer full posterior distributions that describe the range of parameter values consistent with observed data [6]. This probabilistic representation allows uncertainty to be explicitly quantified and propagated into predictions.

Hierarchical Bayesian models extend this framework by allowing parameters to vary across individuals and items while sharing statistical strength at the population level [3] [6]. Such models are particularly well suited to educational data, where each learner

typically generates only a limited number of observations while broader patterns exist across the learner population.

Recent advances in computational inference have made the practical application of hierarchical Bayesian models increasingly feasible. Algorithms such as Hamiltonian Monte Carlo [7] and its adaptive variant, the No-U-Turn Sampler [4], enable efficient exploration of high-dimensional posterior distributions. Modern probabilistic programming frameworks, including `PyMC` [10], provide convenient tools for specifying and estimating such models.

Among recent studies, the work of Nioche et al. [8] is particularly relevant. Their artificial teacher framework demonstrated that incorporating a principled model of human learning and forgetting can significantly improve review scheduling compared with approaches that ignore memory dynamics. The present work builds on this idea by introducing a hierarchical Bayesian formulation of the memory model, allowing both learner-specific and item-specific variability to be represented while maintaining calibrated uncertainty over model parameters.

Chapter 3

Mathematical Model

3.1 Overview

This project models human memory as a probabilistic process in which the probability of recalling an item depends on the time since it was last reviewed and the number of previous repetitions. The model is based on the assumption that memory strength decreases over time but is reinforced through repeated exposure.

The objective of the model is to estimate the probability that a user successfully recalls an item at a given time.

3.2 Memory Decay Model

The probability of recall is modeled using an exponential decay function:

$$p = \exp\left(-\alpha(1-\beta)^{n-1}\Delta t\right) \tag{3.1}$$

where:

- p is the probability of recall.
- α is the forgetting rate parameter.
- β represents the repetition effect.
- n is the number of repetitions.
- Δt is the time elapsed since the last review.

This formulation captures two key aspects of memory.

First, the parameter α controls the rate at which memory decays. Larger values of α correspond to faster forgetting.

Second, the parameter β represents the effect of repetition. As the number of repetitions increases, the effective decay rate decreases. This reflects the well-established observation that repeated exposure strengthens memory.

3.3 Observation Model

In the learning process, recall outcomes are observed as binary variables. A value of 1 indicates successful recall, while a value of 0 indicates failure.

The recall outcome is modeled as a Bernoulli random variable:

$$\omega \sim Bernoulli(p) \tag{3.2}$$

where:

- ω is the observed outcome.
- p is the recall probability.

This probabilistic formulation allows the model to account for variability in recall performance.

3.4 Frequentist Parameter Estimation

In the initial stage of the project, parameters were estimated using maximum likelihood estimation. The likelihood function is defined as:

$$\mathcal{L}(\alpha, \beta) = \prod p^{\omega}(1-p)^{1-\omega} \tag{3.3}$$

The corresponding log-likelihood is:

$$\log \mathcal{L} = \sum \left[\omega \log p + (1-\omega)\ log(1-p)\right] \tag{3.4}$$

Parameter values were obtained by maximizing this log-likelihood function.

Although this approach provides parameter estimates, it has several limitations. In particular, it does not provide information about uncertainty and can produce unstable estimates when data is limited.

These limitations motivate the use of Bayesian methods, which are described in the next chapter.

Chapter 4

Bayesian Hierarchical Model

4.1 Motivation for Bayesian Modeling

The frequentist formulation described in the previous chapter provides point estimates of memory parameters by maximizing the likelihood function. While this approach is computationally simple, it suffers from several limitations. In particular, maximum likelihood estimation does not quantify uncertainty in parameter estimates and may lead to unstable solutions when observations are limited.

Human learning data typically exhibits substantial variability across individuals and learning items.Therefore a modeling framework capable of capturing such variability while sharing statistical strength across observations is required.*Bayesian inference* provides a principled approach to addressing these challenges by treating model parameters as random variables rather than fixed quantities.

Under the Bayesian framework, inference is performed by estimating the posterior distribution of parameters given observed data.

4.2 Bayesian Formulation

Let $\mathcal{D}$ denote the obseved data set consisting of recall outcomes. Bayesian inference aims to compute the posterior distribution

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})} \tag{4.1}$$

where:

- $p(\mathcal{D} \mid \theta)$ is the likelihood,
- $p(\theta)$ represents prior beliefs about parameters,
- $p(\mathcal{D})$ is the marginal likelihood.

Since the marginal likelihood is generally intractable, posterior inference is performed using sampling-based methods described later.

4.3 Prior Specification

The memory model contains two primary priors:

- forgetting rate α
- repetition effect β

Both parameters are constrained. The forgetting rate must remain positive, while the repetition parameter lies within the interval $(0, 1)$.

To ensure these constraints while maintaining favorable sampling geometry, transformed parameterizations are adopted.

4.3.1 Initial Prior Specification

In the initial Bayesian formulation, conjugate priors were considered for model parameters. The forgetting rate was modeled using an exponential prior,

$$\alpha \sim \text{Exponential}(\lambda_\alpha),$$

ensuring positivity of the parameter. The repetition effect was modeled using a Beta prior,

$$\beta \sim \text{Beta}(a, b),$$

4.3.2 Reparameterization for Efficient Inference

Although the conjugate formulation provides intuitive prior choices, preliminary experiments revealed poor sampling behavior when using Hamiltonian Monte Carlo methods. In particular, bounded parameters and hierarchical scale interactions produced unfavorable posterior geometry leading to sampling divergences.

To improve inference stability, transformed parameterizations were introduced. which naturally constrains the parameter to the unit interval.

4.3.3 Forgetting Rate

Instead of placing a prior directly on α, a logarithmic transformation is introduced.

$$\log \alpha \sim \mathcal{N}(0, 1) \tag{4.2}$$

and

$$\alpha = \exp(\log \alpha) \tag{4.3}$$

This transformation guarantees positivity and improves numerical stability during inference.

4.3.4 Repetition Effect

Similarly, the repetition effect is defined through a sigmoid transformation:

$$\beta_{\text{raw}} \sim \mathcal{N}(0, 1) \tag{4.4}$$

$$\beta = \sigma(\beta_{\text{raw}}) \tag{4.5}$$

where $\sigma(\cdot)$ denotes the logistic sigmoid function. This ensures that β remains bounded between zero and one.

4.4 Hierarchical Structure

Memory behavior varies across users and learning items. To model this variability, hierarchical structure is introduced.

Each user and item is associated with latent parameters governing forgetting and repetition dynamics.

4.4.1 Graphical Model Representation

To provide a structured representation of the proposed hierarchical memory model, the probabilistic dependencies between variables are illustrated using a graphical model representation. The diagram summarizes the generative assumptions underlying the learning and forgetting process, including user-specific and item-specific variability as well as the observation mechanism governing recall outcomes.

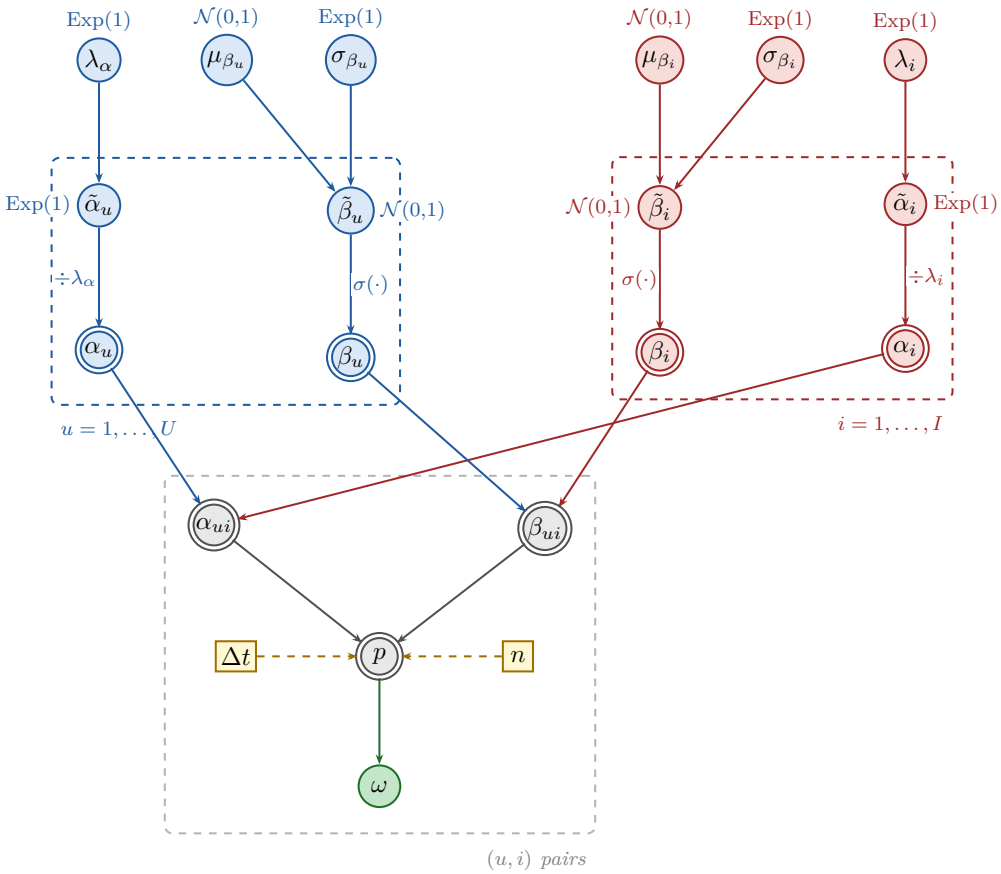

Figure 4.1: Plate diagram of the hierarchical Bayesian memory decay model. The recall probability is computed as $p = \exp\left(-\alpha_{ui}(1-\beta_{ui})^{\max(n-1,0)} \cdot \Delta t\right)$, with $\omega \sim \text{Bernoulli}(p)$. Shaded nodes are observed; double circles are deterministic nodes; squares denote fixed inputs.

Figure 4.1 presents the probabilistic graphical model of the proposed hierarchical memory framework. At the population level, hyperparameters govern the distributions of user- and item-specific parameters, capturing natural variability across learners and learning materials. For each user u, the forgetting rate α_u controls how quickly a memory decays, while repetition sensitivity β_u reflects the extent to which the learner benefits from repeated review. Each item i possesses analogous parameters α_i and β_i, representing its intrinsic difficulty and memorability. At the interaction level, these parameters are deterministically combined to form effective quantities α_{ui} and β_{ui}, which determine the recall probability through an exponential forgetting function modulated by elapsed time Δt and repetition count n. The observed recall outcome ω_{ui} is modeled as a Bernoulli random variable whose success probability is given by the predicted recall probability. Plate notation indicates replication of this generative process across users, items, and interaction events.

This graphical representation provides a compact summary of the assumed generative process and serves as the foundation for the Bayesian inference procedure described in the following section.

4.4.2 User-Level Parameters

For users indexed by u,

$$\alpha_u = \frac{\alpha_u^{raw}}{\lambda_\alpha} \tag{4.6}$$

where

$$\alpha_u^{raw} \sim \text{Exponential}(1) \tag{4.7}$$

$$\lambda_\alpha \sim \text{Exponential}(1) \tag{4.8}$$

This formulation allows users to exhibit heterogeneous forgetting behavior while sharing population-level structure.

For repetition effects,

$$z_u \sim \mathcal{N}(0, 1) \tag{4.9}$$

$$\beta_u = \sigma(\mu_{\beta_u} + \sigma_{\beta_u} z_u) \tag{4.10}$$

with hyperpriors

$$\mu_{\beta_u} \sim \mathcal{N}(0, 1) \tag{4.11}$$

$$\sigma_{\beta_u} \sim \text{Exponential}(1) \tag{4.12}$$

4.4.3 Item-Level Parameters

An analogous structure is defined for items indexed by i:

$$\alpha_i = \frac{\alpha_i^{raw}}{\lambda_\alpha} \tag{4.13}$$

$$z_i \sim \mathcal{N}(0, 1) \tag{4.14}$$

$$\beta_i = \sigma(\mu_{\beta_i} + \sigma_{\beta_i} z_i) \tag{4.15}$$

This enables the model to capture variation in item difficulty.

4.4.4 Interaction-Level Parameters

Recall probability depends on the interaction between a user and an item. Interaction parameters are constructed by combining user and item effects:

$$\alpha_{ui} = \frac{\alpha_u + \alpha_i}{2} \tag{4.16}$$

$$\beta_{ui} = \frac{\beta_u + \beta_i}{2} \tag{4.17}$$

These interaction parameters define personalized memory dynamics.

4.4.5 Generative Model

The complete generative process can be summarized as follows:

1. Sample population-level hyperparameters,
2. Sample user-specific parameters,
3. Sample item-specific parameters,

4. Compute interaction parameters,

5. Compute recall probability.

$$p = \exp(-\alpha_{ui}(1 - \beta_{ui})^{n-1} dt)$$

6. Generate recall outcome

$$\omega \sim Bernoulli(p)$$

4.5 Non-Centered Parameterization

Hierarchical Bayesian models often suffer from pathological posterior geometry known as *the funnel problem.* In centered parameterizations, small values of scale parameters force latent variables into narrow regions of parameter space, leading to inefficient exploration by Hamiltonian Monte Carlo samplers.

To address this issue, a non-centered parameterization is adopted:

$$\theta = \mu + \sigma z, \quad z \sim \mathcal{N}(0, 1) \tag{4.18}$$

This reparameterization decouples local parameters from global scale parameters and significantly improves sampling efficiency.

The adoption of non-centered parameterization eliminated sampling divergences observed in preliminary experiments and enabled stable posterior inference.

4.6 Posterior Inference Objective

The final inference task consists of estimating the joint posterior distribution:

$$P(\alpha_u, \alpha_i, \beta_u, \beta_i, \lambda_\alpha, \mu_\beta, \sigma_\beta \mid \mathcal{D}) \tag{4.19}$$

Due to the complexity of this distribution, analytical solutions are not available, motivating the use of Markov Chain Monte Carlo methods described in the following chapter.

Chapter 5

Posterior Inference and Sampling

5.1 Inference Objective

The hierarchical Bayesian model introduced in the previous chapter defines a joint probability distribution over observed recall outcomes and latent parameters governing memory dynamics. The goal of inference is to estimate the posterior distribution.

$$p(\Theta \mid \mathcal{D}) \tag{5.1}$$

where Θ denotes all latent variables and hyperparameters of the model and $\mathcal{D}$ represents observed recall data.

Due to the nonlinear structure of the memory model and the presence of hierarchical dependencies, the posterior distribution does not admit a closed-form analytical solution. Consequently, approximate inference methods are required.

5.2 Markov Chain Monte Carlo Methods

Posterior inference is performed using Markov Chain Monte Carlo (MCMC) methods. These methods construct a Markov chain whose stationary distribution corresponds to the target posterior distribution.

Given sufficiently long sampling, expectations with respect to the posterior distribu-

tion can be approximated using Monte Carlo estimates:

$$\mathbb{E}[f(\Theta)] \approx \frac{1}{S}\sum_{s=1}^{S} f(\Theta^{(s)}), \tag{5.2}$$

where $\Theta^{(s)}$ denotes samples drawn from the Markov chain.

Traditional MCMC algorithms such as Random-Walk Metropolis often perform poorly in high-dimensional hierarchical models due to slow exploration of parameter space. For this reason, gradient-based sampling methods are preferred.

5.3 Hamiltonian Monte Carlo

Hamiltonian Monte Carlo (HMC) improves sampling efficiency by incorporating gradient information of the log-posterior distribution. Instead of performing random local proposals, HMC introduces auxiliary momentum variables and simulates Hamiltonian dynamics.

Let θ denote model parameters and define the potential energy:

$$U(\theta) = -\log p(\theta \mid \mathcal{D}) \tag{5.3}$$

An auxiliary momentum variable r is introduced with kinetic energy

$$K(r) = \frac{1}{2} r^T r \tag{5.4}$$

The Hamiltonian is then defined as

$$H(\theta, r) = U(\theta) + K(r). \tag{5.5}$$

Sampling proceeds by simulating the dynamical system

$$\frac{d\theta}{dt} = \nabla_r H \tag{5.6}$$

$$\frac{dr}{dt} = -\nabla_\theta H \tag{5.7}$$

These dynamics allow proposals to move efficiently through regions of high posterior probability while maintaining high acceptance rates.

5.4 No U-Turn Sampler (NUTS)

Selecting appropriate trajectory lengths in HMC is challenging in practice. The No-U-Turn Sampler (NUTS) automatically adapts trajectory length during sampling by preventing trajectories from reversing direction in parameter space.

NUTS eliminates the need for manual tuning of leapfrog steps and enables efficient exploration of complex posterior distributions. For this reason, NUTS was employed as the primary inference algorithm via the `PyMC` probabilistic programming framework.

5.5 Posterior Geometry in Hierarchical Models

Despite the advantages of HMC, *hierarchical Bayesian models* often exhibit challenging posterior geometries. A common issue arises from strong dependence between local parameters and global scale parameters.

Consider a simplified hierarchical structure:

$$\theta_i \sim \mathcal{N}(0, \sigma^2) \tag{5.8}$$

$$\sigma \sim p(\sigma) \tag{5.9}$$

When σ approaches small values, all θ_i must lie near zero, creating a narrow region of high probability. Conversely, large values of σ allow wide dispersion of θ_i.

This produces a characteristic geometry known as *the funnel distribution*, consisting of a wide region connected to a narrow neck.

Such geometries are difficult for HMC to explore because step sizes suitable for wide regions become unstable inside narrow regions, leading to divergent transitions.

5.6 The Funnel Problem in Memory Modeling

In preliminary versions of the proposed memory model, exponential and Beta priors were directly combined with hierarchical scale parameters. Empirical sampling diagnostics revealed divergent transitions and poor chain mixing.

These issues originated from coupling between user-level parameters and shared hyperparameters controlling variability across users and items. The resulting posterior

exhibited funnel-like geometry, significantly degrading sampling efficiency.

5.7 Non-Centered Parameterization

To address these difficulties, a non-centered parameterization was adopted.

Instead of sampling parameters directly from:

$$\theta_i \sim \mathcal{N}(\mu, \sigma^2) \tag{5.10}$$

the model introduces auxiliary variables

$$z_i \sim \mathcal{N}(0, 1) \tag{5.11}$$

and defines

$$\theta_i = \mu + \sigma z_i \tag{5.12}$$

This transformation reduces posterior dependency between θ_i and σ, improving geometric structure and enabling more efficient exploration by HMC.

In the proposed memory model, this strategy was applied to repetition parameters and hierarchical components governing forgetting dynamics.

5.8 Sampling Diagnostics

Model convergence was assessed using standard MCMC diagnostics:

- Trace plots to evaluate chain mixing,
- Effective sample size estimates,
- Potential scale reduction statistic ($\hat{R}$),
- Divergence monitoring during sampling.

Successful inference was characterized by $\hat{R}$ values close to one and the absence of divergent transitions, indicating reliable posterior exploration.

5.9 Posterior Interpretation

The resulting posterior samples provide distributions over forgetting rates and repetition effects rather than single point estimates. This enables uncertainty-aware predictions of recall probability and supports personalized modeling of learning behavior. Posterior uncertainty plays a crucial role in adaptive teaching systems, allowing decision-making procedures to account for confidence in parameter estimates when scheduling future reviews.

Chapter 6

Implementation

This chapter describes the computational realization of the proposed hierarchical Bayesian memory model introduced in previous chapters. While earlier chapters established the theoretical formulation and inference methodology, the presentchapter focuses on how the artificial teacher framework was implemented in practice. Particular attention is given to model construction, data processing, and the posterior inference pipeline used for learning learner-specific memory dynamics.

6.1 System Overview

The proposed system models human learning as a probabilistic process governed by forgetting dynamics and repetition effects. The implementation translates the mathematical formulation into an executable Bayesian inference pipeline capable of estimating latent memory parameters from observed recall data.

The overall workflow consists of the following stages:

1. preprocessing interaction data,
2. constructing the hierarchical probabilistic model,
3. performing posterior inference using gradient-based sampling,
4. generating recall probability predictions.

Figure 6.1 illustrates the computational workflow of the implemented artificial teacher system.

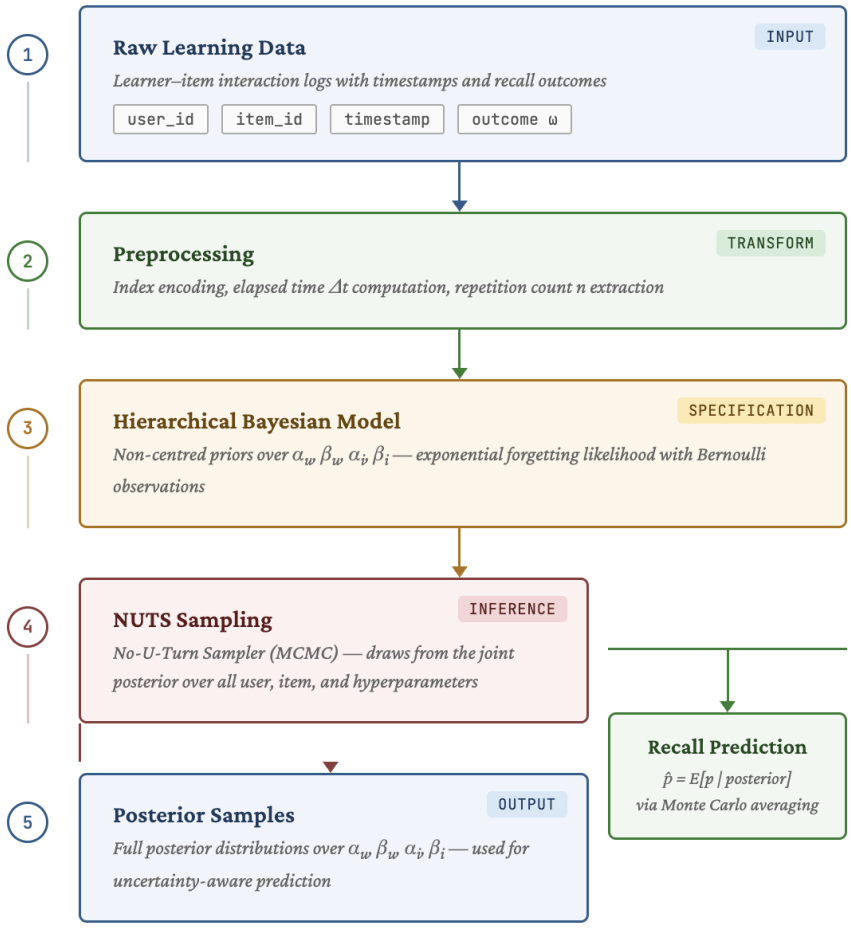

Figure 6.1: Implementation pipeline of the hierarchical memory model, illustrating data preprocessing, probabilistic model construction, posterior inference, and prediction generation.

6.2 Software Framework

The model was implemented in Python using probabilistic programming tools designed for Bayesian inference.

Posterior estimation was performed using the `PyMC` framework, which provides automatic differentiation and efficient implementations of Hamiltonian Monte Carlo and the No-U-Turn Sampler (NUTS). Numerical operations were handled using `NumPy`, while posterior diagnostics and visualizations were generated using the `ArviZ` and `Matplotlib` libraries.

The probabilistic programming paradigm allows the model structure to closely mirror its mathematical formulation, reducing inconsistencies between theoretical specification and implementation.

6.3 Data Representation

Learning interactios were interpreted as tupels:

$$(u, i, n, \Delta t, \omega),$$

where

- u denotes the learner,
- i represents the learning item,
- n corresponds to repetition count,
- Δt denotes elapsed time since last review,
- $\omega \in \{0, 1\}$ indicates recall success.

Categorical user and item identifiers were mapped to integer indices to enable efficient vectorized computation within the probabilistic model.

Time intervals were normalized to ensure numerical stability during exponential decay computation.

6.4 Model Construction

The hierarchical Bayesian model described in Chapter 4 was implemented directly within the probabilistic programming framework.

User-specific and item-specific parameters were defined through hierarchical prior distributions:

$$\alpha_u \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \qquad \beta_u \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2),$$

with analogous definitions for item-level parameters.

Effective interaction parameters were computed as combinations of user and item components,

$$\alpha_{ui} = f(\alpha_u, \alpha_i), \qquad \beta_{ui} = g(\beta_u, \beta_i)$$

which determine recall probability through the exponential forgetting model.

$$p_{ui} = \exp\left(-\alpha_{ui}(1 - \beta_{ui})^{n-1}\Delta t\right)$$

Observed recall outcomes were modeled as:

$$\omega_{ui} \sim \text{Bernoulli}(p_{ui}).$$

This formulation provides personalized estimation of learning behaviour while sharing statistical strength across the population.

6.5 Parametrization Strategy

To ensure stable sampling behaviour, parameters constrained to positive domains were transformed using suitable link functions. In particular forgetting rates were represented in unconstrained space and mapped through exponential transformations.

Furthermore, the non-centered parameterization introduced in Chapter 5 was employed to mitigate posterior funnel geometries. Auxiliary standard normal variables were sampled and transformed according to

$$\theta_i = \mu + \sigma z_i, \qquad z_i \sim \mathcal{N}(0, 1).$$

This reparameterization significantly improved sampling efficiency and reduced divergent transitions during inference.

6.6 Posterior Inference Pipeline

Posterior inference was performed using the No-U-Turn Sampler (NUTS), which automatically adapts trajectory length and step size during Hamiltonian simulation.

Sampling proceeded through the following stages:

1. automatic initialization of model parameters,
2. warm-up (adaptation) phase,
3. Hamiltonian trajectory simulation,
4. posterior sample collection.

Four independent Markov chains were executed to assess convergence reliability.

6.7 Simulation and Training Workflow

Algorithm 6.2 summarizes the overall training procedure used in the implementation.

Training Pipeline

1. Specify ground-truth user and item parameters.
2. Define simulated learner–item interaction tensors.(user indices, repetition counts, elapsed times, and recall outcomes)
3. Construct hierarchical Bayesian model.
4. Initialize sampler.
5. Run NUTS sampling.
6. Store posterior samples.
7. Compute recall predictions.

Figure 6.2: Training workflow of the implemented artificial teacher model.

To evaluate the correctness of posterior inference, synthetic learner–item interaction data were generated using predefined ground-truth parameters. This controlled setup helps in assessment of parameter recovery and verification of the proposed hierarchical memory model prior to deployment on real-world data.

6.8 Computational Considerations

Hierarchical Bayesian models introduce substantial computational requirements due to repeated gradient evaluations during sampling. Therefore, vectorized operations were used wherever possible to reduce computational overhead.

Sampling performance was monitored using convergence diagnostics, including effective sample size and $\hat{R}$ statistics provided by the `ArviZ` framework.

The final implementation balances expressive probabilistic modeling with practical computational feasibility, enabling scalable inference over large collections of learners and learning items.

Chapter 7

Posterior Analysis and Model Behavior

7.1 Overview

Following posterior inference via the No-U-Turn Sampler (NUTS), this chapter shifts emphasis from prediction to interpretation. Rather than asking how accurately the model performs, the more revealing question is what the model has learned about how individual learners forget.

This distinction is central to the Bayesian approach. Unlike conventional machine learning systems, which treat internal parameters as a means to an end, a Bayesian cognitive model regards the posterior distribution itself as the primary output. Each sample drawn via Markov Chain Monte Carlo (MCMC) represents a plausible account of a learner's memory dynamics, and the spread of those samples reflects genuine uncertainty about individual behaviour rather than numerical imprecision.

The analysis presented in this chapter examines four aspects of the fitted model:

- **Convergence behaviour** of the MCMC sampling process;
- **Posterior distributions** of user-specific memory parameters;
- **Variability across learners** and what it reveals about the population;
- **Dependency structure** between inferred forgetting and retention parameters.

7.2 Experimental Setup

Due to the absence of large-scale longitudinal learner data, experiments were performed using a controlled synthetic dataset. User identities, item interactions, repetition counts, elapsed time intervals, and recall outcomes were generated to simulate realistic learning scenarios.

Each interaction record consists of:

- learner index u,
- item index i,
- elapsed time since last review Δt,
- repetition count n,
- observed recall outcome $\omega \in \{0, 1\}$.

Synthetic ground-truth parameters were specified to represent heterogeneous learner memory characteristics. This controlled environment enables qualitative validation of whether the hierarchical Bayesian model can recover underlying behavioural structure from observations.

7.3 Posterior Distribution of User Parameters

The hierarchical model estimates user-specific forgetting parameters α_u and repetition sensitivity parameters β_u. Rather than producing single deterministic values, Bayesian inference yields distributions representing plausible parameter values conditioned on observed interaction data.

Posterior histograms were generated for each learner using samples obtained after convergence of the NUTS sampler.

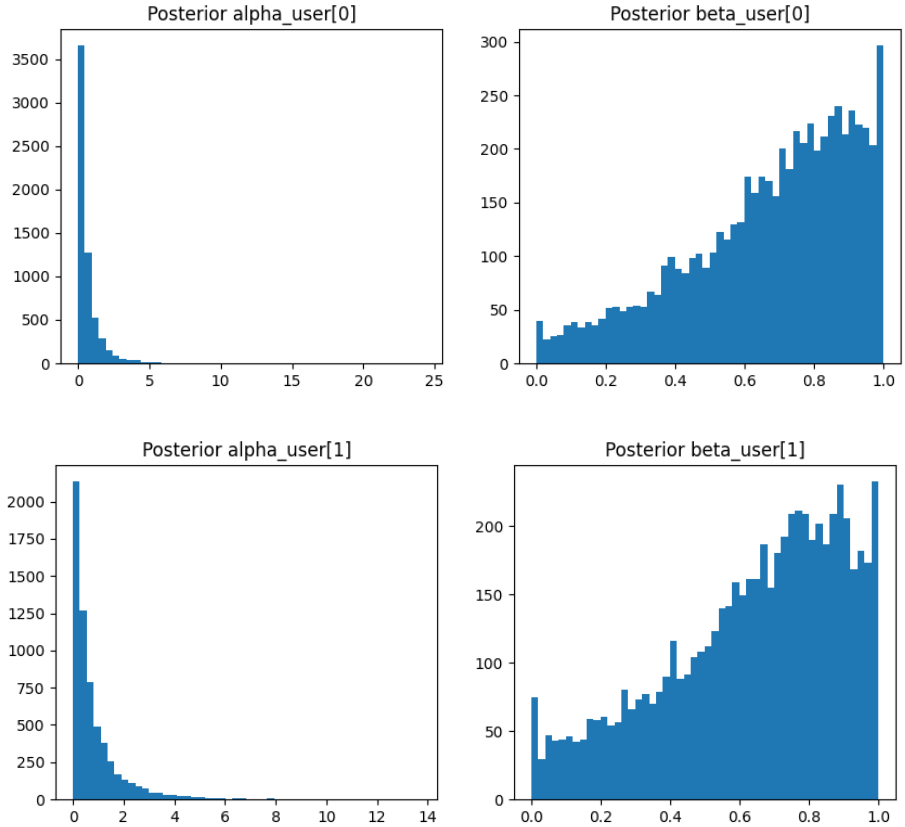

Figure 7.1: Posterior distributions of user-level forgetting (α_u) and repetition (β_u) parameters.

The spread of each distribution reflects uncertainty in parameter estimation. Users exhibiting concentrated posterior mass correspond to well-identified learning behaviour, whereas wider distributions indicate limited observational evidence or higher behavioural variability.

7.4 Sampling Behaviour and Convergence Diagnostics

Reliable Bayesian inference requires that Markov chains adequately explore the posterior distribution. Sampling behaviour was therefore examined using trace plots of inferred parameters across iterations.

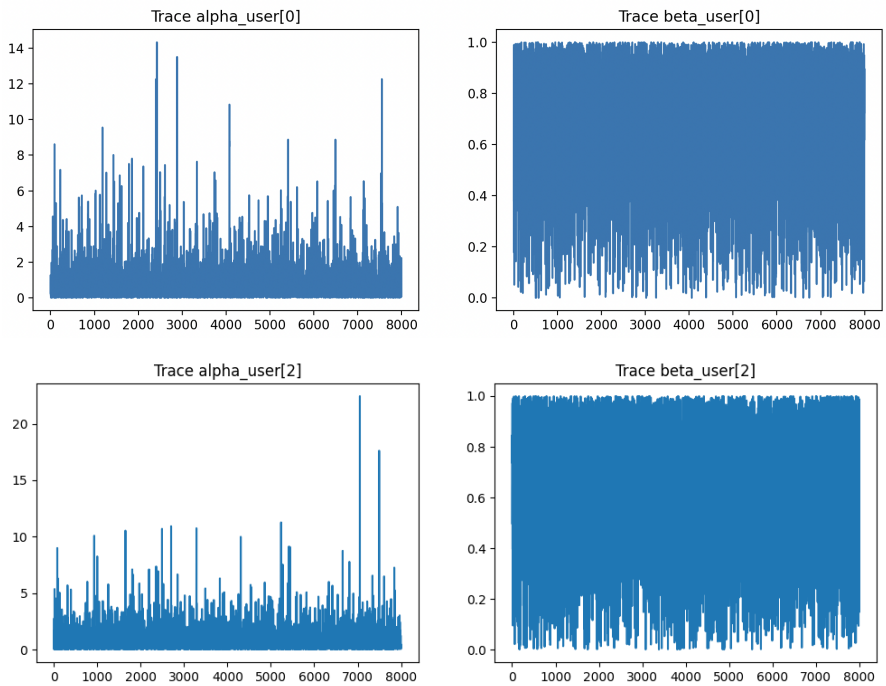

Figure 7.2: Trace plots illustrating sampling trajectories of user parameters across MCMC iterations.

Well-mixed traces without systematic drift indicate stable exploration of the posterior space. The absence of persistent trends suggests convergence of the sampling process and reliable estimation of latent parameters.

7.5 Variability Across Learners

A central motivation for hierarchical modelling is capturing heterogeneity between learners. To analyze inter-user variability, posterior samples of α_u were compared across users.

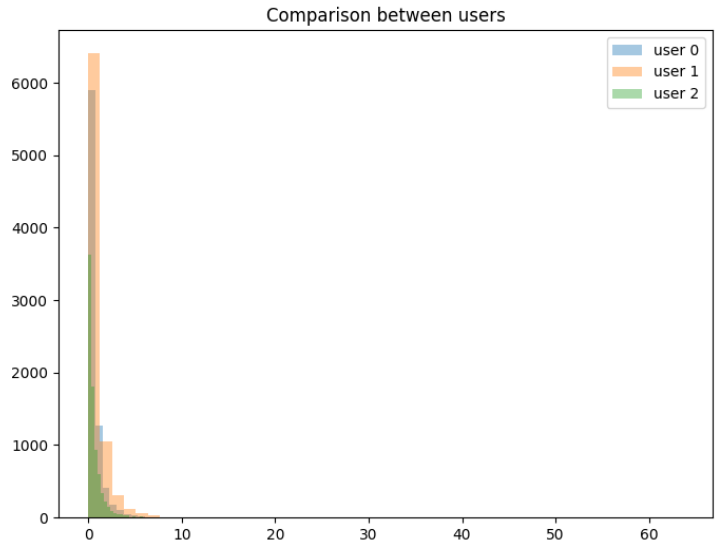

Figure 7.3: Comparison of posterior forgetting-rate distributions across users.

Differences in posterior location reveal individualized forgetting dynamics. Some users exhibit larger forgetting rates, corresponding to faster memory decay, while others demonstrate slower forgetting behaviour.

This result highlights the ability of the artificial teacher model to represent personalized learning characteristics rather than assuming homogeneous memory processes.

7.6 Joint Posterior Structure

Understanding relationships between inferred parameters provides insight into model behaviour. Joint posterior samples of α_u and β_u were visualized using scatter and density plots.

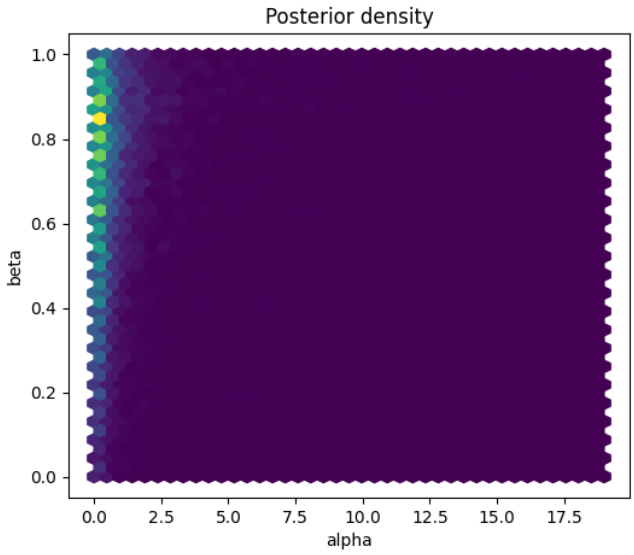

Figure 7.4: Joint posterior density illustrating dependency between forgetting and repetition parameters.

The joint distribution reveals how forgetting rates and repetition effects interact within the inferred memory model. Regions of high density correspond to parameter combinations consistent with observed recall outcomes.

Such analysis is particularly important in hierarchical Bayesian systems, where dependencies between latent variables influence posterior geometry and learning interpretation.

7.7 Interpretation for Artificial Teaching Systems

A central advantage of the Bayesian artificial teacher lies in its ability to reason explicitly under uncertainty. Rather than assigning each learner a single fixed forgetting rate or retention sensitivity, the model maintains posterior distributions over these parameters. These distributions capture not only the most plausible characteristics of a learner's memory, but also the degree of confidence supported by observed interaction data.

This probabilistic representation has direct implications for personalised review scheduling. When posterior distributions become concentrated, the system can make confident predictions about future recall behaviour and schedule repetitions with high precision. Conversely, diffuse posterior distributions indicate limited evidence about a learner's memory dynamics, suggesting that additional observations are required before reliable

scheduling decisions can be made.

Uncertainty therefore becomes an operational signal rather than a statistical artifact. The artificial teacher adapts not only to estimated learner abilities but also to how certain those estimates are. Posterior analysis thus serves a dual purpose: it characterises latent memory processes inferred from data and provides the principled foundation for uncertainty-aware, adaptive teaching strategies.

Chapter 8

Discussion, Limitations, and Future Directions

8.1 Discussion

The experimental results demonstrate that the hierarchical Bayesian framework successfully infers interpretable learner-specific memory representations from interaction data.

Unlike traditional predictive models that optimise accuracy alone, the proposed approach yields probabilistic descriptions of learning behaviour together with principled uncertainty estimates. Posterior variability reflects genuine ambiguity arising from limited observations rather than optimisation noise.

Although experiments were conducted using synthetic data, the observed inference behaviour suggests that the framework scales naturally to real educational environments where learner dynamics are inherently uncertain and heterogeneous.

The results therefore validate the feasibility of Bayesian artificial teaching systems capable of adapting review decisions based on inferred cognitive structure.

8.2 Limitations

Several limitations should be acknowledged. First, experiments rely on synthetic interaction data rather than large-scale real learner datasets. Consequently, quantitative predictive evaluation remains limited.

Second, computational cost associated with MCMC sampling may pose challenges for

real-time deployment in large educational systems. Future work may therefore investigate approximate inference techniques such as variational methods.

Despite these constraints, the present study establishes a foundational probabilistic framework upon which scalable artificial teaching systems can be developed.

8.3 Conclusion

This report presented the development of a hierarchical Bayesian artificial teacher designed to model how learners acquire, retain, and forget information over time. Motivated by the need for personalised and uncertainty-aware systems, the proposed framework integrates probabilistic memory modelling with modern Bayesian inference techniques.

A hierarchical formulation was introduced to capture variability across both learners and learning items. Forgetting behaviour and repetition sensitivity were modelled as latent parameters governed by population-level distributions, enabling statistical strength to be shared across individuals while preserving personalised adaptation.

Posterior inference was performed using Hamiltonian Monte Carlo and the No-U-Turn Sampler (NUTS), allowing efficient exploration of the complex posterior geometry induced by hierarchical dependencies. Particular attention was devoted to addressing sampling difficulties through non-centered parameterization, which substantially improved inference stability.

Experimental analysis demonstrated that the model successfully recovers interpretable learner-specific memory characteristics. Posterior distributions revealed structured variability across learners and provided principled uncertainty estimates rather than single deterministic parameter values. These probabilistic representations form the operational basis of the artificial teacher, enabling adaptive reasoning about when review should occur and how confidently scheduling decisions can be made.

Importantly, the results illustrate a conceptual shift from prediction-driven learning systems toward inference-driven probabilistic models. In this framework, the objective is not merely to predict recall outcomes but to infer latent cognitive processes governing learning behaviour.

The work therefore establishes a coherent probabilistic foundation for artificial teaching systems capable of personalised and uncertainty-aware decision making.

8.4 Future Directions

Several directions can be explored naturally from the present study.

8.4.1 Evaluation on Real Learning Data

The current implementation relies on synthetically generated interaction data to validate inference behaviour under controlled conditions. A critical next step involves deployment on real-world educational datasets containing longitudinal learner interactions. Such evaluation would enable quantitative assessment of predictive performance and practical scheduling effectiveness.

8.4.2 Scalable Approximate Inference

While Hamiltonian Monte Carlo provides high-quality posterior estimates, its computational cost may limit scalability in large-scale learning platforms. Future research may investigate approximate inference approaches such as variational inference or amortized Bayesian methods to enable real-time adaptive teaching.

8.4.3 Richer Cognitive Models

Human learning involves additional factors beyond forgetting and repetition effects, including contextual interference, spacing strategies, and learner fatigue. Incorporating richer cognitive mechanisms within the hierarchical Bayesian framework represents a promising direction for future artificial teacher systems.

8.4.4 Integration with Intelligent Tutoring Systems

Finally, integration of Bayesian memory models into interactive intelligent tutoring environments would allow continuous updating of learner representations during real educational use, bridging the gap between probabilistic modelling and deployed adaptive education technologies.

References

[1] John R Anderson and Lael J Schooler. Reflections of the environment in memory. *Psychological science*, 2(6):396–408, 1991.

[2] Hermann Ebbinghaus. *Memory: A Contribution to Experimental Psychology*. Teachers College, Columbia University, 1885.

[3] Andrew Gelman et al. *Bayesian Data Analysis*. CRC Press, 2013.

[4] Matthew Hoffman and Andrew Gelman. The no-u-turn sampler. *Journal of Machine Learning Research*, 2014.

[5] Robert V Lindsey, Jeffery D Shroyer, Harold Pashler, and Michael C Mozer. Improving students' long-term knowledge retention through personalized review. *Psychological science*, 25(3):639–647, 2014.

[6] Richard McElreath. *Statistical Rethinking*. CRC Press, 2 edition, 2020.

[7] Radford M. Neal. Mcmc using hamiltonian dynamics. *Handbook of Markov Chain Monte Carlo*, pages 113–162, 2011.

[8] Aurélien Nioche, Pierre-Alexandre Murena, Carlos de la Torre-Ortiz, and Antti Oulasvirta. Improving artificial teachers by considering how people learn and forget. In *Proceedings of the 26th International Conference on Intelligent User Interfaces*, pages 445–453, 2021.

[9] Philip Pavlik and John Anderson. The effect of practice on learning. *Cognitive Science*, 2005.

[10] John Salvatier et al. Probabilistic programming in python using pymc3. *PeerJ Computer Science*, 2016.

[11] Piotr Woźniak and Edward Gorzelańczyk. Optimization of repetition spacing in the practice of learning. *Acta neurobiologiae experimentalis*, 54(1):59–62, 1994.